NILABHRO MOJUMDAR

ASPIRING DATA ANALYST

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PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineering graduate with foundational SQL, Excel, MATLAB and Java skills and experience in quantitative engineering research. Applied comparative analysis to evaluate signal processing methods using measurable performance indicators. Interested in data cleaning, KPI reporting, analytical problem-solving and business intelligence.

TECHNICAL SKILLS

Existing: SQL • Microsoft Excel • MATLAB • Java • Flutter • Quantitative analysis • Performance benchmarking

Analytics: SQL joins, CTEs, window functions • Data cleaning • Power BI • Python/Pandas • Dashboarding • Customer analytics

ACADEMIC PROJECTS

Channel Equalization in MIMO Systems | MATLAB • Experimental Analysis

- Investigated signal distortion, noise and interference in wireless channels and compared LMS, Zero Forcing (ZF) and FLANN-LMS adaptive equalization methods.
- Structured comparisons around signal-to-noise ratio (SNR) and bit error rate (BER) to evaluate how methods respond to changing channel conditions.
- Interpreted algorithm performance and trade-offs, identifying FLANN-LMS as effective for adapting to nonlinear patterns and variable channel characteristics.

Performance Analysis of Spatial Modulation in MIMO Systems | Engineering Research

- Studied spatial modulation, where antenna index information is used alongside conventional modulation with one active transmitting antenna at a time.
- Analyzed design trade-offs related to transmission performance and system complexity; synthesized technical findings into an organized comparative study.

Number Guessing Game | Java

- Developed a console-based guessing game using randomized target numbers and iterative user input; explored extensions for rounds and score tracking.
- Applied conditional logic, loops and basic debugging to handle user interactions and game flow.

Retail Sales & Profitability Intelligence | SQL • Excel • Power BI

- Project scope: Model orders, order items, customers and products using SQL joins, CTEs and aggregations; calculate revenue, order count, average order value and category performance. The source does not provide product cost, so gross margin requires a separately documented cost assumption.
- Clean duplicate transactions, missing values and inconsistent categories; reconcile SQL totals against source data and document assumptions and validation checks.
- Create an interactive Power BI dashboard with date/region/category filters, period-over-period comparisons and drilldowns to identify underperforming products and revenue drivers.

Open dataset: Olist Brazilian E-Commerce Public Dataset (Kaggle) — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Customer Segmentation & Retention Analysis | SQL • Python/Pandas

- Analyze transaction-level data using SQL window functions and date logic to calculate recency, frequency, monetary value (RFM), repeat-purchase rate and customer cohorts.
- Use Python/Pandas for exploratory analysis, outlier treatment and segment validation; compare purchasing behavior across high-value, at-risk and newly acquired customers.
- Present actionable retention opportunities through segment-level KPIs and visualizations, distinguishing observed patterns from causal conclusions.

Open dataset: UCI Machine Learning Repository, Online Retail II (Daqing Chen, 2012) — https://archive.ics.uci.edu/dataset/502/online+retail+ii

Operations & Inventory Exception Monitoring | SQL • Excel

- Project scope: Combine daily item-level sales and store-level price data to estimate sales velocity, rolling demand, slow movers and potential stockout exposure; inventory turnover and true stock coverage require stock-on-hand data not supplied in this dataset.
- Use SQL CASE expressions, window functions and rolling averages to flag unusual demand, price changes and potential replenishment exceptions; summarize priority items in an Excel exception tracker.
- Document data-quality checks and recommended replenishment actions, emphasizing transparent thresholds and business impact.

Public dataset: M5 Forecasting — Accuracy (Walmart unit sales, calendar and prices; Kaggle competition) — https://www.kaggle.com/competitions/m5-forecasting-accuracy/data

VOCATIONAL TRAINING

National Aluminium Company (NALCO) — Alumina Refinery | Control & Instrumentation Department

23 May 2022 – 20 June 2022

- Completed vocational training within the Control & Instrumentation (C&I) department of an industrial alumina refinery, gaining exposure to the role of measurement and process control in plant operations.
- Developed contextual understanding of how instrumentation and operational parameters support monitoring, reliability and process decision-making in a manufacturing environment.
- Relevant analytical connection: process measurements can be organized into time-series KPIs, trends and exception reports for operational performance reviews. (Potential application; not claimed as work performed during training.)

EDUCATION

B.Tech, Electronics & Telecommunication Engineering | Veer Surendra Sai University of Technology (VSSUT), Burla | 2020–2024 | CGPA: 6.94/10